

Competitive Benchmark: Reading Support App for Children with Dyslexia

Authors: Rayen Ancamilla, Eduardo Gómez, Raul Manriquez

Course: HUMAN-COMPUTER INTERACTION DESIGN (ICC716 -1)

May 17, 2026

1 00 - Benchmark Summary

1.1 1. Feature Map

Figure 1 presents the distribution and feature mapping of the tools analyzed.

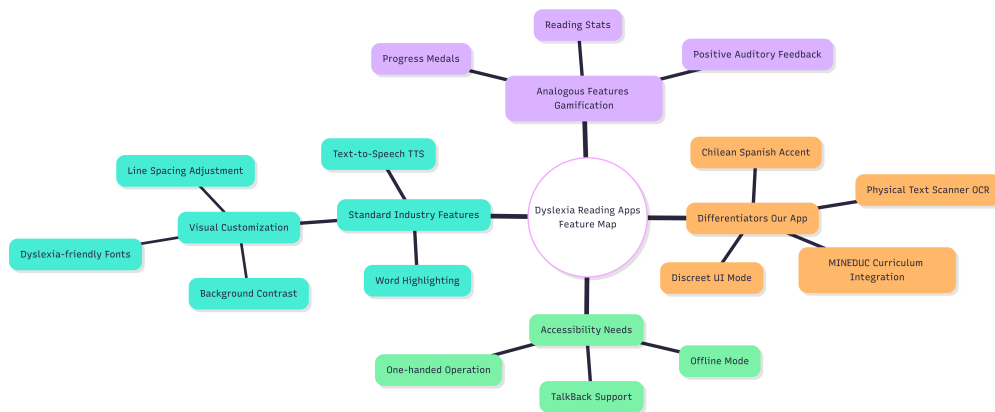


Figure 1: Feature map of competitors.

1.2 2. Synthetic Comparative Table

Analysis Dimension	Voice Dream Reader (Direct)	Google Read Along (Analogous)	Spotify (Design Ref.)	Group Proposal (DyslexiaApp CL)
Target User Profile	Adults and students with SLD.	Young children (5-11 years old) learning to read.	General public, audio listeners.	Children (8-12 years old) with dyslexia in the Chilean school system.
Value Proposition	Highly customizable premium text-to-speech reader.	AI-powered reading tutor with gamification.	The best seamless listening experience.	Discreet app, adapted to Chile and the MINEDUC curriculum.
Navigation	Bottom nav bar (Library, Settings).	Guided flow through a level map (game style).	Bottom nav bar with persistent player.	Bottom nav bar (Library, Scanner, Achievements, Profile). Persistent player.
Accessibility	High (dyslexia fonts, high contrast, VoiceOver).	Medium (large icons, voice prompts, but lacks fine visual settings).	High (car view, high contrast, intuitive UI).	High (Auditory feedback, TalkBack, clean and high-contrast UI).
MINEDUC Integration	No. Texts must be loaded manually.	No. Uses generic short stories.	N/A.	Yes. Pre-loaded library with Chilean school texts.
Chilean Adaptation	Yes, but relies on the OS voice engine (robotic).	No. Generic neutral/Latin American Spanish.	N/A.	Yes. TTS voices with local accent and adapted vocabulary.
Discretion in Class	Low. Utilitarian and complex interface, reveals its special use.	None. Very noisy, invasive gamification.	High. Dark and minimalist interface.	High. Modern and “normal” interface, can be used with headphones without drawing attention.

2 01 - Tool A: Voice Dream Reader (Direct Competitor)

2.1 Base Dimensions

- **Identification:** Voice Dream Reader — iOS/Android — Analyzed May 2026.
- **Target user profile:** College students and adults with dyslexia or visual impairments.
- **Value proposition:** “The most accessible way to read.” Powerful TTS reader adaptable to any format (PDF, EPUB).
- **Main features:** (1) Text-to-Speech (TTS), (2) Synchronous word highlighting, (3) Visual settings (OpenDyslexic, spacing), (4) File importation.
- **Onboarding flow:** Complex. Requests storage permissions immediately and requires a long gesture tutorial to learn how to use the player.
- **Navigation patterns:** Bottom tab navigation. Heavy use of modal menus for visual settings.
- **Visual design and consistency:** Utilitarian, minimalist but slightly outdated. Prioritizes function over aesthetics.

2.2 Domain-Specific Dimensions

- **MINEDUC Integration:** None. The child/parent must scan or find the PDF on their own, generating friction.
- **Chilean Adaptation:** Relies on system voices (Siri/Google TTS). Lacks vocal empathy.
- **Discretion in Class:** Medium-Low. The interface looks like a clinical accessibility tool, which may cause embarrassment for the child in front of peers.

2.3 Strengths and Areas for Improvement

- **Strength 1:** Deep visual customization.
- **Strength 2:** Dual-tone highlighting (highlights the entire line and the specific word being read).
- **Area for Improvement 1 (Heuristic: Error prevention):** Importing complex documents (poorly scanned PDFs) breaks the reader without giving the user a clear error message.
- **Area for Improvement 2 (Heuristic: Flexibility and efficiency of use):** The audio settings menu has too many sliders grouped together (speed, pitch, volume), overwhelming a child user.

2.4 Annotated Screenshots

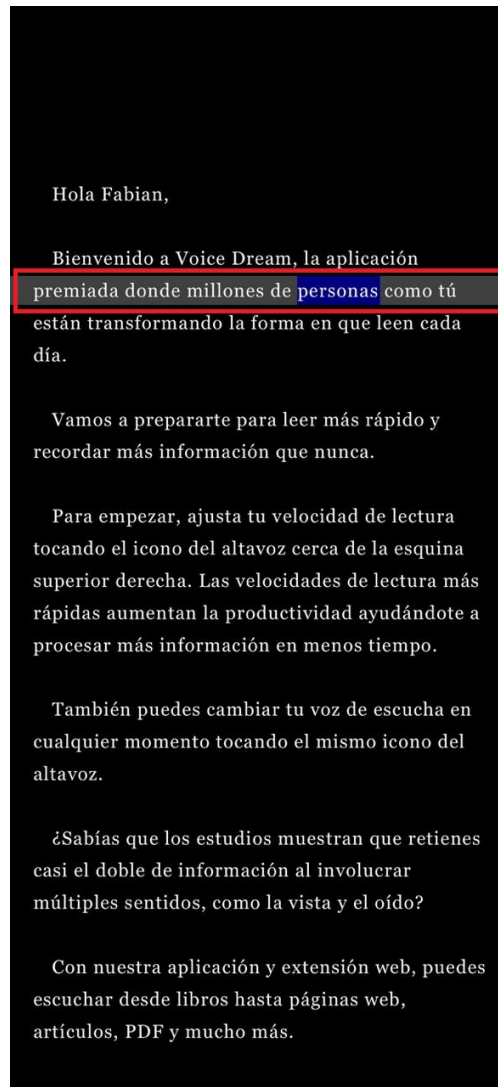


Figure 2: Main reading screen in Voice Dream Reader.

- **Screen/Flow:** Main reading screen.
- **Visual Annotation:** A red box outlining the sentence and the word being read, highlighted in dual-tone.
- **Relevance Comment:** Dual highlighting facilitates visual tracking without getting lost in the paragraph. We will adopt this standard for our app.

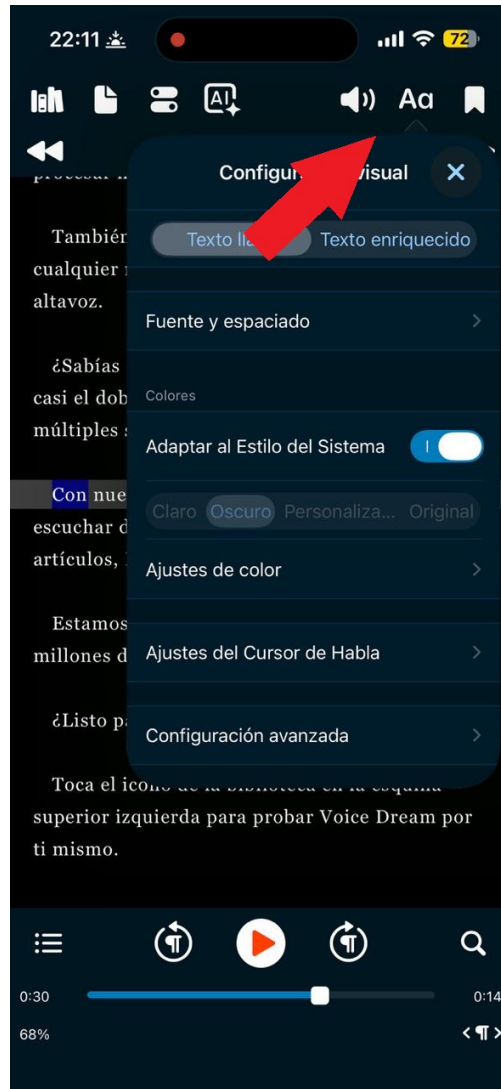


Figure 3: Visual settings menu in Voice Dream Reader.

- **Screen/Flow:** Visual settings menu.
- **Visual Annotation:** An arrow pointing to the dense grouping of typography and audio options and sliders.
- **Relevance Comment:** The cognitive overload of this menu violates minimalist design heuristics. We must simplify these controls so as not to overwhelm children.

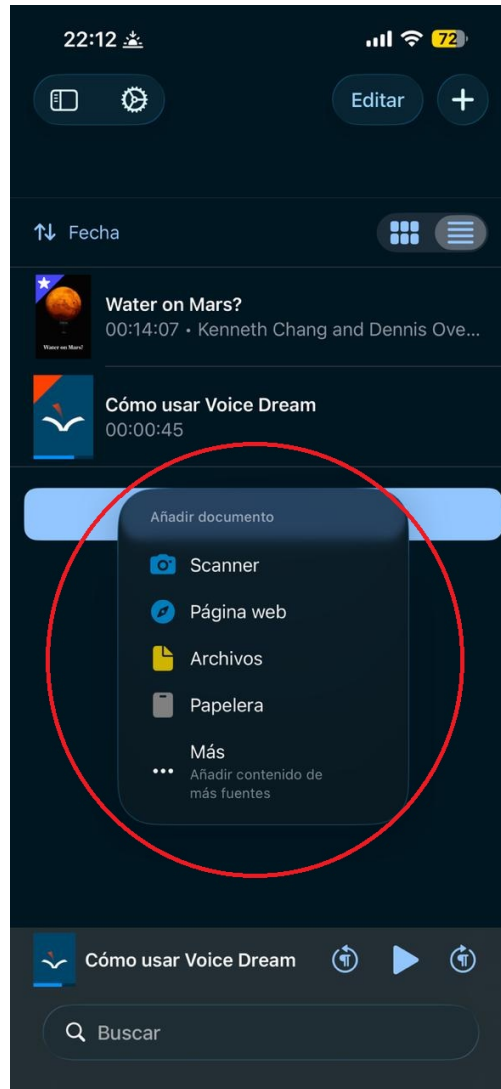


Figure 4: File import flow in Voice Dream Reader.

- **Screen/Flow:** File import flow (Onboarding).
- **Visual Annotation:** A circle around the manual “Add Document” button.
- **Relevance Comment:** The need to manually search for and import files adds too much friction. This justifies our decision to include a pre-loaded library with MINEDUC texts.

3 02 - Tool B: Google Read Along (Analogous Competitor)

3.1 Base Dimensions

- **Identification:** Google Read Along — Android — Analyzed May 2026.
- **Target user profile:** Young children (5-11 years old) learning to read or improving their fluency.
- **Value proposition:** “Helps kids learn to read with an interactive voice assistant (Diya).”
- **Main features:** (1) Active user listening, (2) Real-time correction, (3) Story library, (4) Minigames and rewards.
- **Onboarding flow:** Very friendly and fast. The assistant does a quick reading test to adjust the level.
- **Navigation patterns:** Non-linear navigation based on a spatial map.
- **Visual design and consistency:** Playful, vibrant colors, consistent children’s illustrations.

3.2 Domain-Specific Dimensions

- **MINEDUC Integration:** None. Only contains short children’s stories. Not useful for studying school guides or the required reading plan.
- **Chilean Adaptation:** Uses generic neutral/Latin American Spanish.
- **Discretion in Class:** None. Forces the child to read out loud to their device, which exposes our UX Persona to social stress.

3.3 Strengths and Areas for Improvement

- **Strength 1:** The visual positive reinforcement system (animated stars) fosters self-esteem without interrupting the reading flow.
- **Strength 2:** Reading interface free of heavy cognitive distractions; there are no complex buttons.
- **Area for Improvement 1 (Heuristic: User control and freedom):** The user is heavily limited by the flow imposed by the voice assistant; free navigation is difficult.
- **Area for Improvement 2 (Heuristic: Match between system and the real world):** For 10 to 12-year-old children, the aesthetic is excessively childish.

3.4 Annotated Screenshots

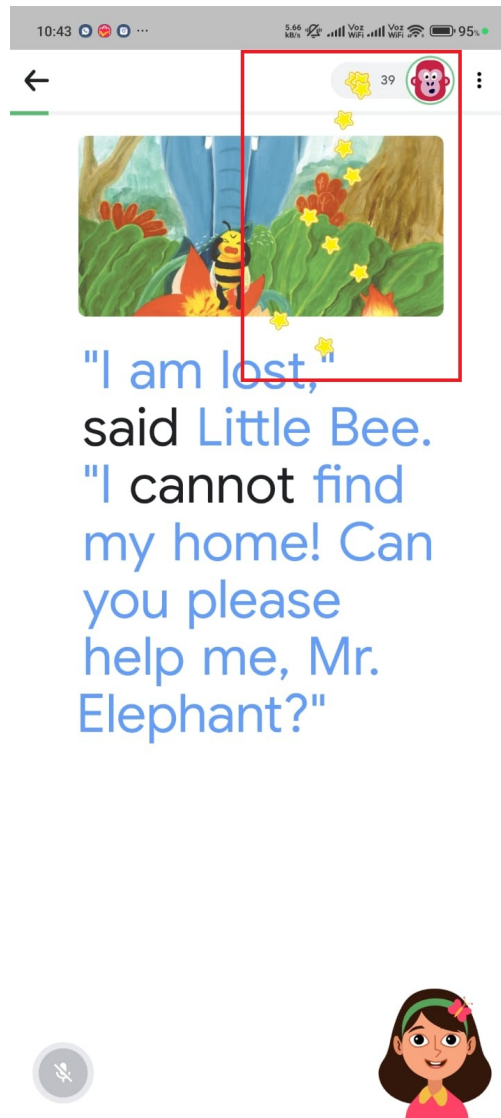


Figure 5: Positive reinforcement screen in Google Read Along.

- **Screen/Flow:** Positive reinforcement screen.
- **Visual Annotation:** A red box around the animated reward stars.
- **Relevance Comment:** This visual micro-interaction boosts self-esteem without interrupting the flow. We will adapt this pattern for our statistics.

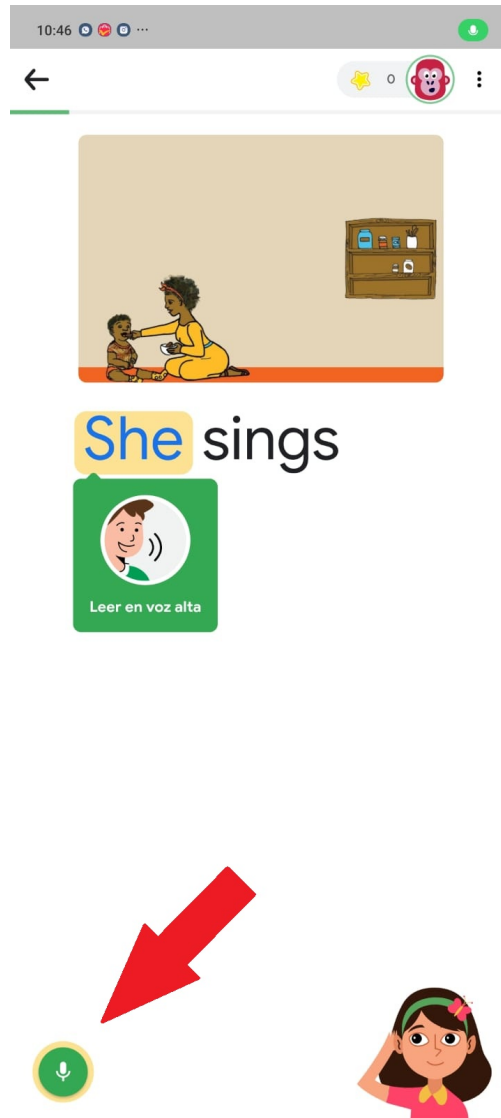


Figure 6: Assistant interface waiting for user audio.

- **Screen/Flow:** Reading assistant waiting for user audio.
- **Visual Annotation:** An arrow pointing to the active microphone icon.
- **Relevance Comment:** Forcing the user to read aloud aggravates social anxiety for introverted children like Sofía. We will reject this pattern.

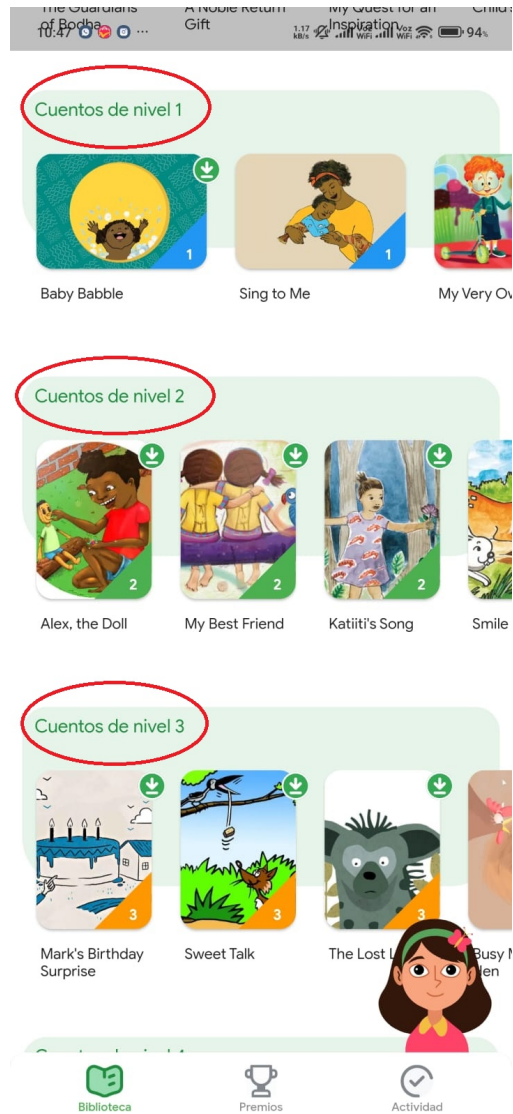


Figure 7: Navigation map (Home screen) in Google Read Along.

- **Screen/Flow:** Navigation map (Home screen).
- **Visual Annotation:** A circle grouping the asymmetrical “path” structure of the stories.
- **Relevance Comment:** This playful map is inefficient for searching school texts. We will use a catalog structure so users like Tomás can study autonomously.

4 03 - Tool C: Spotify (Design Reference)

4.1 Base Dimensions

- **Identification:** Spotify — Android / PC — Analyzed May 2026.
- **Target user profile:** General users who consume audio.
- **Value proposition:** Frictionless listening of content, library organization, and real-time lyrics tracking.
- **Main features:** (1) Persistent mini-player, (2) Real-time synchronized lyrics, (3) Offline downloading.
- **Onboarding flow:** Standard registration focused on quick selection of artists.
- **Navigation patterns:** Classic Bottom Nav Bar. Heavy use of the minimized player floating above the rest of the interface.
- **Visual design and consistency:** Default Dark Theme, minimalism, high contrast, and highly legible thick sans-serif typography.

4.2 Domain-Specific Dimensions

- **Discretion in Class:** Spotify is the universal standard for a “normal app”. If our app borrows its visual codes (minimalist bottom player, dark theme), the child will feel they are using a modern app, achieving 100% discretion.

4.3 Strengths and Areas for Improvement

- **Strength 1:** The “Synchronized Lyrics” pattern illuminates exactly the phrase playing. It is the standardized equivalent to the text highlighting (TTS) needed by children with dyslexia.
- **Strength 2:** The persistent mini-player in the bottom bar allows the user to continue browsing the library without stopping the current audio.
- **Area for Improvement 1 (Heuristic: Visibility of system status):** In the lyrics view, if the dynamic background takes a very light color, the white text loses contrast, making reading difficult.
- **Area for Improvement 2 (Heuristic: Error prevention):** Secondary buttons (shuffle, repeat) have small hit-boxes, which could generate accidental touches.

4.4 Annotated Screenshots



Figure 8: Home screen with active playback in Spotify.

- **Screen/Flow:** Home screen with active playback.
- **Visual Annotation:** A box highlighting the persistent bottom mini-player.
- **Relevance Comment:** The mini-player allows continuous navigation without losing text control. It provides the discreet and modern design feel our UX Personas seek.

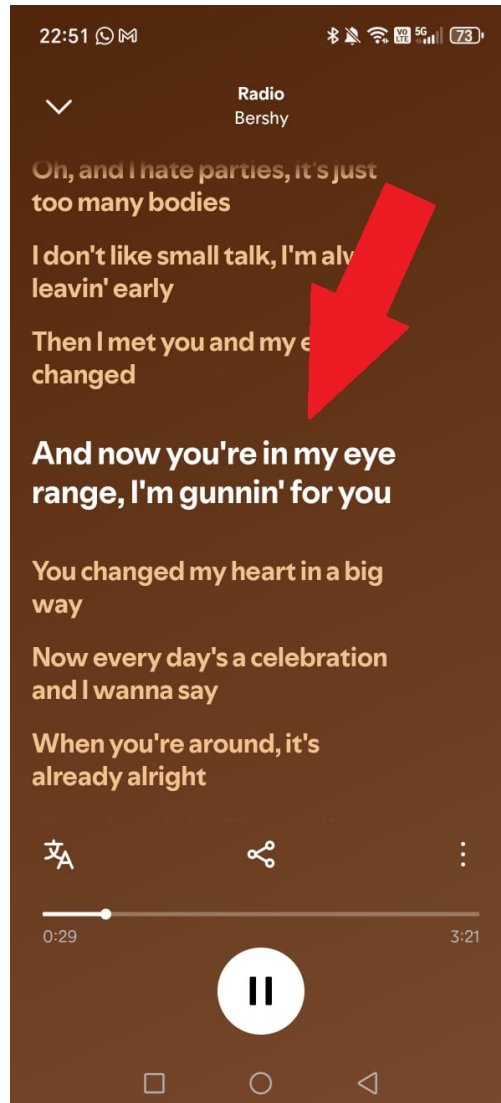


Figure 9: Synchronized lyrics screen in Spotify.

- **Screen/Flow:** Synchronized Lyrics screen.
- **Visual Annotation:** An arrow indicating the bright white active line contrasting with the dark/transparent upcoming text.
- **Relevance Comment:** This synchronized highlighting is highly familiar and reduces visual stress. It will be our standard for the text-to-speech (TTS) tool.

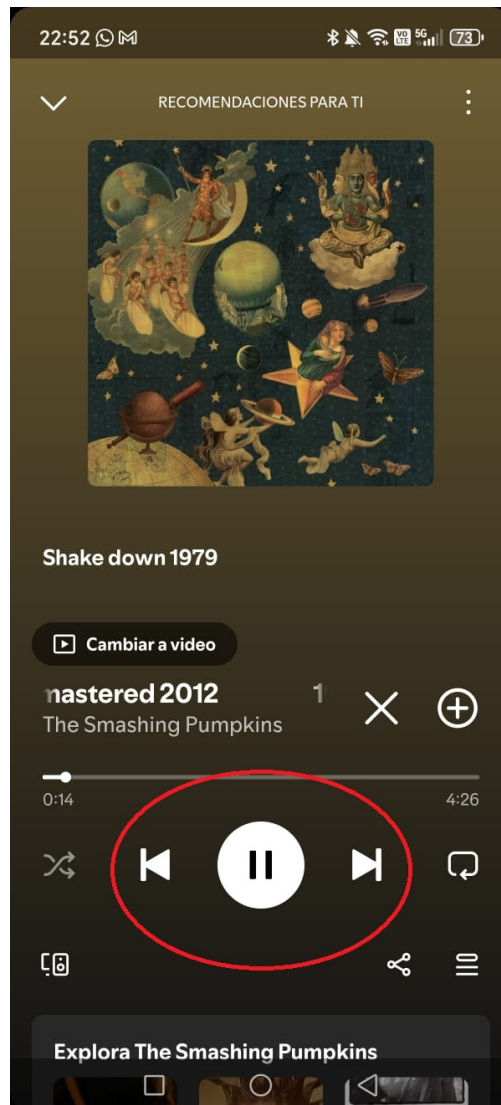


Figure 10: Main playback controls in Spotify.

- **Screen/Flow:** Main playback controls.
- **Visual Annotation:** A circle around the central buttons (Play, Pause, Next).
- **Relevance Comment:** The generous hit-boxes and high contrast of these buttons are our direct accessibility reference to prevent touch errors.

5 Findings and Design Decisions

Based on the Comparative Table, the Feature Map, and the detailed analysis of Phases 1 and 2, we have made the following direct decisions to nourish our Value Canvas and Information Architecture diagram:

1. **Adoption of Pre-Loaded MINEDUC Library:** No competitor solves the parents' pain point ("not having time to search and load books"). Our architecture (see *diagrama app.jpg*) includes "Curricular Texts" as the first level of the Library.
2. **Rejection of the "User Reading Out Loud" Pattern:** Unlike Google Read Along, we will not force the user to read into the microphone. Sofia is introverted and fears reading in public. Our app will be a Passive Reader (TTS) for the user.
3. **Adoption of Music Player Patterns (Spotify Reference):** To meet the requirement of a "discreet and modern design" and for the child to integrate "without feeling different", we will adopt a persistent bottom player and a dark visual design (Dark Mode). If the peers of Tomás or Sofia see their screen, it will look like they are listening to a podcast or music, not using a clinical tool.
4. **Simplification of Visual Settings:** Unlike Voice Dream Reader which cognitively overloads the user, our "Visual Settings" will be limited to 3 pre-tested base configurations of high contrast, OpenDyslexic/Arial typography, and size, delegating the complexity to the parents through the Profile.